

Time & Work — Remaining Variants

The Last 2–5%: Rarer TCS Digital Phrasings + Extra Trigger-Word Tricks

Aptitude Grind

Your first three booklets (01-fundamentals, 02-fast-tricks-and-shortcuts, 03-question-pattern-catalog) cover essentially all high-frequency TCS Ninja + Digital Time & Work questions. This file adds the small remainder: ten less-common variants and seven trigger-word tricks that occasionally show up, mostly on the Digital-level paper. Treat this as a short top-up, not a new foundation — every method below still reduces to the core ideas in 01-fundamentals.tex.

Category K — Rare But Seen Variants

K1. Working for x Days, Then Leaving

Phrasing

“A can finish a job in 20 days. A works for 6 days and then leaves. B finishes the remaining work alone in 21 days. In how many days can B alone finish the whole job?”

Method: Completed work $\rightarrow$ remaining work $\rightarrow$ solve. A’s 6-day share = $6/20 = 3/10$; remaining = $7/10$. If B does that $7/10$ in 21 days, B’s full-job time = $21/(7/10) = 30$ **days**.

K2. Working Together, Then One Leaves

Phrasing

“A and B together can finish a job in 8 days. They work together for 5 days, then A leaves. B finishes the rest alone in 6 more days. In how many days can B alone do the whole job?”

Method: Joint work $\rightarrow$ remaining work $\rightarrow$ single worker. Combined rate = $1/8$; 5-day joint share = $5/8$; remaining = $3/8$. B does $3/8$ in 6 days $\Rightarrow$ B’s full-job time = $6/(3/8) = 16$ **days**.

K3. Fraction of Work Completed by Each Person

Phrasing

“A completed $3/5$ of a work in 12 days, then left. B completed the remaining work in 6 days. Find the time each would take to complete the full work alone, and the time taken if they had worked together from the start.”

Method: Scale each person’s fraction up to a full job first — A: $12/(3/5) = 20$ days alone; B: $6/(2/5) = 15$ days alone. Then combine as usual: together = $(20 \times 15)/(20 + 15) = 60/7$ days.

K4. Reduced Efficiency Due to a Mid-Job Slowdown**Phrasing**

“A can finish a job in 15 days working normally. After 5 days, A starts working at half the usual efficiency for the rest of the job. How many total days does A take?”

Method: Split into two phases with two different rates — normal rate first, halved rate second; don't try to average the rates. First 5 days at $1/15/\text{day}$ complete $1/3$; remaining $2/3$ at half rate ($1/30/\text{day}$) takes 20 more days $\Rightarrow$ **Total = 25 days**.

K5. Repeating Break / Rest Cycle**Phrasing**

“A works for 5 days and rests on the 6th (repeating). A alone can finish a job in 24 days of actual work. How many total days (including rest days) will the job take?”

Method: Treat the rest day as adding to the calendar, not to the work. 4 full 6-day cycles cover $4 \times 5 = 20$ working days in $4 \times 6 = 24$ calendar days; 4 working days remain, needing 4 more calendar days $\Rightarrow$ **Total = 28 calendar days**. Always separate “days worked” from “days elapsed” for this variant.

K6. Machine Breaks Down and Is Repaired**Phrasing**

“A machine can complete a job in 12 hours. It runs for 3 hours, then breaks down and is under repair for 2 hours (doing no work), then resumes until the job is finished. How long does the whole job take, start to finish?”

Method: Split into phases exactly like a pipe/machine problem, but insert a zero-work phase for the downtime. Work done in first 3 hours = $3/12 = 1/4$; remaining $3/4$ at the same rate needs 9 more running hours $\Rightarrow$ **Total elapsed = 3 + 2 (repair) + 9 = 14 hours**.

K7. Wrong Time Assumption / Efficiency-Loss Question**Phrasing**

“A estimated finishing a job in 10 days at a certain daily rate, but actually took 12 days at a lower daily rate. By what percentage did A's actual daily efficiency fall short of the estimate?”

Method: Efficiency is inversely proportional to time (01-fundamentals.tex §6). Estimated rate $\propto 1/10$, actual rate $\propto 1/12$. Percentage shortfall = $(1/10 - 1/12)/(1/10) \times 100 = 16.67\%$.

K8. Workforce Increases by a Percentage (Not Efficiency)

Phrasing

“20 workers can complete a job in 18 days. If the number of workers is increased by 20%, in how many days will the job be completed?”

Method: $\text{Workers} \times \text{Days} = \text{Constant}$ (same worker-day logic as 02-fast-tricks-and-shortcuts.tex §3, applied to headcount instead of a mid-job change). New workers = $20 \times 1.2 = 24$; new days = $(20 \times 18)/24 = 15$ **days**. Keep this distinct from an *efficiency* increase — here the per-person rate is unchanged, only headcount changes.

K9. Unit-Production Phrasing (Bottles, Units, Items per Hour)

Phrasing

“Machine A produces 200 bottles per hour, Machine B produces 150 bottles per hour. Working together, how long will they take to produce 7000 bottles?”

Method: Identical to a combined-work question — treat “bottles” as the job. Combined rate = $200 + 150 = 350$ bottles/hour; time = $7000/350 = 20$ **hours**. Don’t be thrown by the absence of the word “work” — any per-unit-time output phrasing (bottles, pages, units, bricks) uses the same rate-addition method.

K10. Mixed Skilled / Unskilled Workers

Phrasing

“3 skilled workers and 5 unskilled workers can complete a job in 6 days. A skilled worker is twice as efficient as an unskilled worker. If 4 skilled workers alone were to do the job, how many days would they take?”

Method: Convert every worker to one common unit first. Let unskilled rate = 1 unit/day $\Rightarrow$ skilled rate = 2 units/day. Combined rate = $3(2) + 5(1) = 11$ units/day; total work = $11 \times 6 = 66$ units. 4 skilled workers’ rate = $4 \times 2 = 8$ units/day $\Rightarrow$ time = $66/8 = 8.25$ **days**.

Extra Trigger-Word Tricks

These are quick mental reflexes to pair with the ten variants above — the same “see the word, do the step” style as 02-fast-tricks-and-shortcuts.tex.

“...then leaves / then joins”

Completed work $\rightarrow$ Remaining work. Never solve the whole job in one pass; split into phases at the moment the workforce changes.

“...twice / thrice / half as efficient”

Convert straight to a rate ratio (2:1, 3:1, 1:2). Never convert to a day ratio first — that inversion step is where most errors happen.

“...every Nth day / alternate days / repeating cycle”

Think in whole cycles, not day-by-day. Compute work done per cycle, find how many full cycles fit, then solve only the small leftover.

“...ratio of times / ratio of days is given”

Immediately flip it: time ratio and efficiency ratio are always inverses of each other.

25% / 40% / 75% (any clean percentage)

Convert to a fraction before touching the arithmetic ($25\% \rightarrow 1/4$, $40\% \rightarrow 2/5$, $75\% \rightarrow 3/4$). Fractions combine faster than decimals in every one of these variants.

All given times are whole numbers

Default to the LCM Method from `02-fast-tricks-and-shortcuts.tex` §1 — it applies just as well to K3, K9, and K10 as it does to the core combined-work questions.

Pipes

Keep fill/production as + and empty/downtime as a 0-rate phase, and always track *elapsed time* separately from *working time* when a repair or rest period is involved (K5, K6).

What This File Deliberately Skips

Per the original review, these remain **not worth prepping for TCS** and are intentionally excluded here as well: multi-page algebraic worker puzzles, hour-by-hour efficiency changes, irregular A/B/C/D rotation schedules, infinite pipe on/off sequences, nonlinear efficiency models, and minimum-time optimization/scheduling problems. These belong to CAT/SSC CGL-level prep, not TCS NQT.

Updated Priority Order

With this file added, the full priority order across all four booklets becomes:

1. **A1** — Two people working together (base formula)
2. **B1** — Efficiency ratio (“X times as efficient”)
3. **C1/C2** — Someone joins or leaves midway
4. **K1/K2** — Working for x days then leaving / together-then-one-leaves (same family as C1/C2)

5. **D1** — Men/Women/Children equivalence
6. **E1/E2** — Pipes and cisterns, including a pipe closing partway
7. **F1** — Alternate-day working
8. **K5** — Repeating break/rest cycle (same family as F1)
9. **G1** — Wages split by work share
10. **H1** — Machine/Printer/Robot work
11. **K6** — Machine breaks down and is repaired (same family as H1)
12. **I1** — Percentage of work completed
13. **K3** — Fraction of work completed by each person (same family as I1)
14. **K9** — Unit-production phrasing (bottles/pages/units)
15. **J1/J2** — Variable efficiency
16. **K4** — Reduced efficiency due to mid-job slowdown (same family as J1/J2)
17. **K8** — Workforce increases by a percentage
18. **K10** — Mixed skilled/unskilled workers
19. **K7** — Wrong time assumption / efficiency-loss question (lowest frequency, Digital only)

At this point the coverage is effectively complete for TCS NQT. The highest-value next step is pure repetition: solve 40–60 mixed questions across all four booklets until each pattern is recognized on sight, since TCS rewards speed of recognition over depth of math.